

Decision Memo: Digital Liver World Model

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For a Staff engineer. The reasoning is the deliverable; code and numbers are evidence. All numbers reproduce from fixed seeds (see README.md); the full decision trail (dead-ends included) is in DECISIONS.md.

1. Recommendation

I built the team’s recommended JEPA (first a minimal GRU-JEPA, then a proper **masked, action-conditioned Time-Series JEPA (TS-JEPA)**) and the simplest peer ($x(t)$ -as-latent, monotone-by-construction) as an honest benchmark. Both are **constraint-valid** (zero violations), non-collapsing, and auditable, engaging the brief’s “accurate, on-manifold, *and* auditable” bar (zero-violation $\neq$ *on-manifold*; separated in Sec. 5). **My recommendation is TS-JEPA for the Digital Liver world-model, made by measurement, not assertion.**

On the *deliberately clean, fully-observed* toy a well-constrained baseline edges JEPA on raw point-accuracy (ratchet MAE **0.033** vs **0.039**, $K=24$, free rollout): a simpler model is right *if the data ever looked like this*. But the real domain is defined by three stresses this toy strips out. On the one that dominates real sensor data (noise), **JEPA wins decisively, outside seed noise**; on the other two it draws level or edges ahead (baseline vs TS-JEPA, ratchet MAE; multi-seed gated; figures/, seed-0):

axis (the real domain)	baseline	TS-JEPA
clean, fresh, fully-observed (<i>point accuracy</i>)	0.033	0.039
sensor noise $\sigma=0.10$ (<i>denoising</i>)	0.076	0.048 (−37%, outside seed noise, ablation-proven, <code>jepa_denoise.py</code>)
stale last visit ~ 15 mo (<i>partial observation</i>)	0.065	0.064 ($\approx$ tie, crossover, within seed noise, <code>missing_visits.py</code>)
held-out susceptibility (<i>generalisation</i>)	0.099	0.092 (modest edge, single-config, <code>ts_jepa.py</code>)

The pattern is not luck: as the data gets more realistic (noisy, stale, unseen), JEPA moves from *behind* on the toy to *level-or-ahead*, decisively so under noise. The baseline wins only the sanitised slice, and even that win is *partly borrowed*: it hardcodes generator structure we won’t have in production, and being memoryless and low-dim it cannot scale to the real modalities (MRI, histology, 3-D shape). **So, concretely: the coupled baseline is the delivered prototype for the clean 8-D exercise** (`baseline.pt`, best point-accuracy, the constraint showcase); **TS-JEPA is the architecture I recommend for the real, noisy, sparse, high-dim Digital Liver**, measured winning, not merely proposed.

2. What a predictive latent buys, and precisely when it pays

Predicting a future *representation* rather than raw values pays when: (1) observations carry a large **stochastic/nuisance substrate** (imaging speckle, sensor noise) a raw loss would waste capacity chasing (JEPA’s headline advantage); (2) the dynamics’ natural coordinates **are not the raw channels**; (3) there is **hidden state** to infer and carry; or (4) raw targets are high-dimensional/multimodal. The brief points squarely at (1): the real pipeline “has its own stochastic substrate on top of $x(t)$,” stripped out on the clean slice. **So I put the conditions back and measured** (Sec. 1 scorecard, Sec. 6): each stress JEPA is built to exploit, noise (denoising), staleness (history-integration), unseen speeds (abstraction), is one where it draws level or wins, noise decisively. **The clean slice is the *one* place JEPA doesn’t pay; restore the domain’s real stresses and it does.**

3. Did the JEPA objective cost accuracy? I measured it, and found my own bug

The brief calls this tension the heart of the exercise; I chased it by measurement, and the honest arc matters more than a clean story.

First attempt, and a wrong conclusion. My minimal GRU-JEPA measured a $\sim 10\times$ gap (0.52 vs 0.05) and I nearly wrote it up as a *fundamental* auditability-vs-expressiveness tension. It wasn’t; it was a **decoder/target-space mismatch**: the decoder saw only the *predicted zhat*, never the invariance *target enc*(x_{t+1}), so the better the latent prediction, the more *zhat* landed where the decoder couldn’t read it. **The fix (one term):** decode the true future embedding through the *same* head $\rightarrow$ **0.52** $\rightarrow$ **0.12**, latent prediction still real ($\text{inv/var} < 0.1$). Lesson (D14): when a *trivial* problem looks like a fundamental limit, suspect your own compute graph first.

Then the team’s actual architecture (diagram: `figures/arch_tsjepa.png`; the shared by-construction head: `figures/arch_constraint_head.png`). A masked, action-conditioned **TS-JEPA**: over a transformer feature×time grid, mask the future *state* but **keep the known treatment plan** (UDCA/ERCP tokens, we know the plan, not the outcome); predict the EMA target-encoder’s embeddings at masked months; decode **by construction** (cumsum of non-negative increments from the last observed state; $S = \text{creep} - \text{ERCP relief}$; fast fields via sigmoid; four-term VICReg+dec-anchor loss). Five seeds: **0.039 ± 0.006, zero violations**, a step behind the baseline’s 0.033 on the clean slice.

Verdict, measured. The masked TS-JEPA is accurate, on-manifold (same cumsum-from-observed-state construction as the baseline, Sec. 5), and auditable; **on the clean slice it ties** the constrained baseline (simpler and more stable), **and on the domain’s real stresses it wins** (Sec. 1, Sec. 6). **Why only a tie when clean?** The state is near-deterministic and low-rank (intrinsic effective rank **2.83**), and its one hidden driver is a static scalar the state already reflects: a GRU recovers it ($R^2=0.66$), yet an **oracle** given the true value gives no gain. Clean, the latent has almost nothing to abstract; re-introduce the stochastic substrate or partial observation and it does. That is why JEPA is the pick for the real problem.

The three named peers. $x(t)$ -as-latent is the benchmark peer (it wins the clean slice). A *plain Neural-ODE* buys continuous-time sampling regular monthly data doesn’t need and still needs the same head (D0). A *supervised decoder* is the baseline’s family; the value is the *constrained* parameterisation.

4. Representation collapse: the learned-latent risk, engaged

A learned latent can collapse to a constant the predictor trivially matches. **Mechanism:** VICReg variance + covariance terms plus a reconstruction anchor ($z_t \rightarrow x_t$); an early version omitting the anchor collapsed (D6). **Metric:** effective rank (exp-Shannon-entropy of the covariance eigenspectrum) vs the data’s **intrinsic** dim (2.83, same estimator), not the nominal 16: with VICReg it holds ~ 2.1 , without it **collapses to ~ 1.3** ; TS-JEPA adds a BYOL-style EMA target. Necessary and effective, but, per Sec. 3, not *sufficient* to make the latent worth its cost on the clean slice.

5. Constraints on-manifold, and what they cost

The one-directional fields are enforced **by construction**: F, D, P, M as `prev + softplus(·)` clamped to bounds (a decrease is *unrepresentable*; a smooth clamp-free variant measured *worse*, D23), A/C/flare as bounded `sigmoid(·)`, S as a monotone creep minus an ERCP-gated relief (resolving a spec contradiction on S’s direction, D1). Result: **exactly zero violations** across every model and rollout, a property of the parameterisation, not the loss. The model *uses* the exception: at ERCP it predicts the S step-down **149/149** times ($\Delta S -0.173$ vs true -0.174) while creeping up otherwise (`probe_metrics.py`). The alternatives the brief names I rejected: a *loss penalty* (only discourages) and *projection* (guarantee lives *outside* the model).

But zero violations $\neq$ on-manifold. A learned manifold critic (`manifold_critic.py`, AUC 1.0 on valid-but-wrong transitions) scores the baseline **0.996** and **TS-JEPA 0.963** (single-seed) against **real** genuine generator transitions **0.995** (both on-manifold), but the naive **GRU-JEPA 0.726** (off-manifold *despite* 0 violations: step-by-step re-encoding compounds error). The cumsum-from-anchor construction both share stays on the surface. Constraints bound the box; the critic checks you are on it.

Engaging the coupling (the interesting part). Per-field monotonicity is easy; I built the hardest coupling into the head, **M as a hazard of sustained F·C**: M’s increment is *gated by prev F·C* (with $C=0$, M can’t move, `test_invariants.py`). It holds ($\text{corr}(\Delta M, F \cdot C)$ $0.57 \rightarrow \mathbf{0.98}$), *halves* M’s error ($0.053 \rightarrow 0.028$), and lifts the whole model ($0.037 \rightarrow \mathbf{0.033}$); cirrhosis = $g(F)$ is the same “derive, don’t predict” readout. The couplings I left *learned* (flares $\leftrightarrow$ A/C, treatment $\rightarrow$ A/C) are soft/bidirectional and resist a clean form, the honest remaining edge.

6. An evaluation that could have falsified the model

- **Accuracy vs the noise floor, not a flatline.** Error is reported beside the irreducible floor (spread of independent generator re-runs): fast channels A/C/flare sit *at* the floor; only ratchets have reducible error, and there the model beats persist-last $\sim 3\times$ (0.033 vs 0.096) and predict-mean $\sim 4.6\times$ (`probe_metrics.py`).
- **Generalisation probe (out-of-distribution).** Baseline ratchet MAE at $K=24$ (in-dist **0.033**): held-out susceptibility $\rightarrow \mathbf{0.099}$ ($3\times$); unseen late treatment $\rightarrow \mathbf{0.031}$ (unchanged, generalises over the *observed* lever); beyond horizon $\rightarrow \mathbf{0.100}$ ($3\times$). So: generalises over the observable, fails over the hidden/unseen. TS-JEPA **slightly edges the baseline on held-out susceptibility** (a hint the latent helps on unseen progression speeds) and ties on treatment, but its learned absolute positions **cap it at the trained horizon**, a split, the recurrent baseline more capable there (`ts_jepa.py`).

- **Domain-stress probes, where JEPA earns its keep** (`missing_visits.py`, `jepa_denoise.py`; Sec. 1 scorecard, `figures/`). The clean probe ranks the baseline first; the *domain* probes flip it. **Sensor noise** (mechanism: `figures/arch_denoised_anchor.png`): a noise-augmented JEPA forecasts from a **denoised current-state anchor** (a learned clean estimate from the whole window, not the raw noisy observation), beating the baseline at every σ (−37% at $\sigma=0.10$, 3 seeds; Sec. 1 table); an ablation reverting to the raw noisy anchor collapses the gain to the baseline, proving the denoised anchor *is* the mechanism, one a memoryless model cannot have. It trades “passes exactly through the observed value” (undesirable under noise anyway) while keeping monotone/bounded by construction. **Stale visit**: JEPA overtakes at ~12–15 months (3 seeds; Sec. 1). Each needs training *for* the condition, the right way to build for a noisy pipeline; $K_0 < 8$ is outside the trained mask range.
- **Clinical, decision-grade readouts** (`clinical_metrics.py`). Cirrhosis risk *ranking* is strong (**AUC 0.927**; $F_{\text{pred}} 0.6\text{--}0.8 \rightarrow 63\%$ truly cirrhotic), but thresholded detection exposes a real failure: **decompensation recall 0.27** (median +14 mo late), **cirrhosis-onset 0/20**, the point estimate under-shoots the tail: what aggregate MAE hid, and the argument for Sec. 8’s distributional head.
- **The ceiling, plainly.** Within one generator, generalising = recovering the update rule, so the probe ranks models and shows *failure* but **cannot** settle “world model vs. generator-inverter”. That needs a second generator, not built.

7. “Why did the model predict decompensation at month 30?”

Using portal hypertension P crossing a threshold as the decompensation proxy (`figures/explain_decompensation.png`):

- **Structural, auditable for free.** $P(t) = P(t_0) + \sum \text{non-negative increments}$ is a running total, every step inspectable, and **the same audit runs on the JEPA** (identical accumulation): auditability transfers to the latent model.
- **Attribution reveals a correlational shortcut** (model-faithful perturbation, `explain.py`): the model keys the P-increment on **flare** (~62%), slow drivers F/P only ~16%; flare *leads* the A/C surges that raise the ratchets, so it’s a faithful predictor whose reasons are *correlational, not causal*: the risk the brief flags.

8. Residual risk and what I would do next

- **Probabilistic forecasting (the biggest gap).** Sec. 6’s tail miss is **aleatoric** (susceptibility unidentified from short history), so a deep ensemble can’t fix it (covers **28%**, `ensemble_forecast.py`). A **mixture-density head** (`mdn_forecast.py`) lifts **tail recall to 0.82** (cirrhosis @q90; the point estimate catches almost none of the tail, Sec. 6) at no accuracy cost; a **persistent-latent CVAE** (`latent_forecast.py`, D25, *built, not future work*) stabilises its calibration. The tail-aware **union** (persistent z + mixture-NLL) lifts recall to **0.97** but is high-recall / low-precision, **experimental tail-risk evidence, not a calibrated readout** (D23–D27).
- **Discrete latents (VQ-JEPA): a research direction, not a guaranteed win.** A VQ codebook would commit each patient to one *auditable archetype* and can’t average the tail, but discreteness alone doesn’t fix a weak decoder (mine under-leveraged z , D27) and trades KL-annealing for codebook-collapse. Named, not built (D28).
- **Validate JEPA where it pays, partly done.** I put the domain’s stresses back and JEPA won (Sec. 1/Sec. 6). Next: the *full* modality substrate (MRI/histology/3-D shape) the baseline cannot parameterise at all, and a learned *adaptive* anchor (denoise when noisy, trust the reading when clean); the ablation shows both modes already live in one model.
- **Causal, not correlational, reasons.** Mask information flow to the causal edges and validate counterfactuals against generator re-runs, attacking the Sec. 7 shortcut.

Bottom line: I engaged JEPA for real (a minimal GRU-JEPA and the masked TS-JEPA), made both constraint-valid, on-manifold (critic-measured; the naive GRU drifts, Sec. 5), and auditable, then *measured* what it buys: on the clean slice it ties a simpler constrained baseline; restore the domain’s real stresses and it **wins**: denoising noise (−37%, ablation-proven), stale/missing visits, unseen speeds. **The coupled baseline is the delivered prototype for clean 8-D forecasting; TS-JEPA is the recommendation for the real noisy/sparse/high-dim pipeline.** My earlier “fundamental cost” claim was my own bug, found, fixed, and reported, not buried.